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 Studierichting: Informatica
 Oefeningenreeks 4A nr. 39.6

$$\begin{aligned}
 I &= \int \frac{4x+7}{(x^2+2x+2)^3} dx \\
 &= 2 \int \frac{\cancel{2x+2}}{(x^2+2x+2)^3} \frac{1}{\cancel{2x+2}} d(x^2+2x+2) + 3 \int \frac{1}{(x^2+2x+2)^3} dx \\
 &= \frac{\cancel{-2}}{(x^2+2x+2)^2} + 3 \int \frac{1}{(x^2+2x+2)^3} dx \\
 z &= x+1 \text{ en } dx = dz \text{ en } x^2+2x+2 = (x+1)^2+1 \\
 &\Rightarrow \int \frac{1}{(x^2+2x+2)^3} dx = \int \frac{1}{(z^2+1)^3} dz \\
 &= M_3(z) = \frac{3}{8} \operatorname{Bgtan} z + \frac{3z}{8(z^2+1)} + \frac{z}{4(z^2+1)^2} + C \\
 &\Rightarrow I = \frac{-1}{(x^2+2x+2)^2} + 3M_3(x+1) + C \\
 &= \frac{-1}{(x^2+2x+2)^2} + \frac{9}{8} \operatorname{Bgtan}(x+1) + \frac{9(x+1)}{8(x^2+2x+1)} + \frac{3(x+1)}{4(x^2+2x+1)^2} + C \\
 &= \frac{9}{8} \operatorname{Bgtan}(x+1) + \frac{9(x+1)}{8(x^2+2x+2)} + \frac{3x-1}{4(x^2+2x+2)^2} + C
 \end{aligned}$$

References